

PRODUCT DATA CLEANING & PREPARATION USING EXCEL

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Course Name: AI-Driven Data Analytics

Institute: ENTRi

Introduction:

- This project focuses on cleaning and preparing a product dataset using Microsoft Excel. Real-world datasets often contain missing values, inconsistent formatting, duplicate entries, and unstructured text fields. These issues must be resolved before performing meaningful analysis.
- The objective of this assignment is to apply essential Excel data-cleaning techniques such as handling missing values, correcting inconsistencies, removing duplicates, splitting and merging data, formatting values, and applying conditional formatting. These operations improve the accuracy, readability, and usability of the dataset.

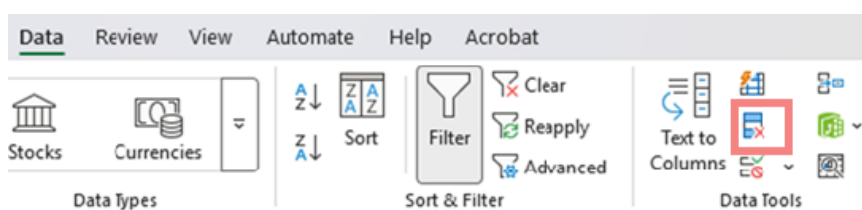
Purpose:

- The purpose of this project is to demonstrate practical data-preprocessing skills using Excel. The tasks include identifying missing values, standardizing text formats, correcting category errors, restructuring columns, and applying formatting rules.
- By completing these tasks, the dataset becomes clean, consistent, and ready for further analysis or reporting.

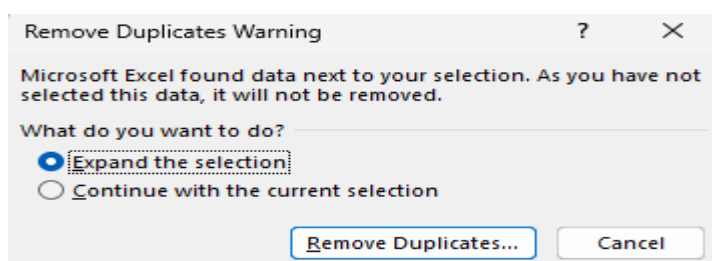
Assignment Tasks and Completed Work

3.1 Removing Duplicates:

- 1. Select the column "Product ID"
- 2. Go to Data --> Data Tools-->Select Remove Duplicate
- 3. Select --> Expand the selection ---> Select all the Tab (Product ID, Product Name, Brand Name, Price, Quantity, Category)
- 4. Select all Tab --> Remove duplicate---> Removed 3 Duplicate value



Select Highlighted Option
"Remove Duplicates"



3.2 Handling Missing Values:

- Checked for missing values in the **Price** column using COUNTBLANK.
- Found **3 blank cells** (J5, J19, J28).
- Replaced missing prices with the **average price** of the column.
- Checked for missing values in the **Category** column.
- Found **5 blank cells**.
- Replaced missing categories with "Unknown".

Task Comments	Output Values	Formula Used
Check for missing values in the 'Price' column.	3	=COUNTBLANK(D2:D36)
How would you handle products with missing price information? Price column is found to blank	298 (J5, J19, J28)	=IF (ISBLANK(D2), AVERAGE(\$D\$2:\$D\$32), D2)
Check for missing values in the 'Category' column.	4	=COUNTBLANK(F2:F36)
How would you handle products with missing Category information? Category column is found to blank	Unknown	=IF (ISBLANK(G2), "UNKNOWN", G2)

3.3 Correcting Inconsistent Data:

- Identified inconsistent text formats in **Product Name**.
- Cleaned using **PROPER**, **TRIM**, and **CLEAN** functions.
- Identified typos in **Category** column (e.g., "Electroni"). → Sorted the column, located incorrect entries, and corrected them using **Find & Replace**. → Replaced **5 incorrect values** with "Electronics".

Task Comments	Output Values
Inconsistent Text formats present in the "Product Name"	Refer Highlighted cells in Column H

Formula Used
=PROPER(TRIM(CLEAN(B2)))

Task Comments	Output Values
Identify any typos present in the "Category" column.	Refer Highlighted cells F6:F10 in Column F

Search

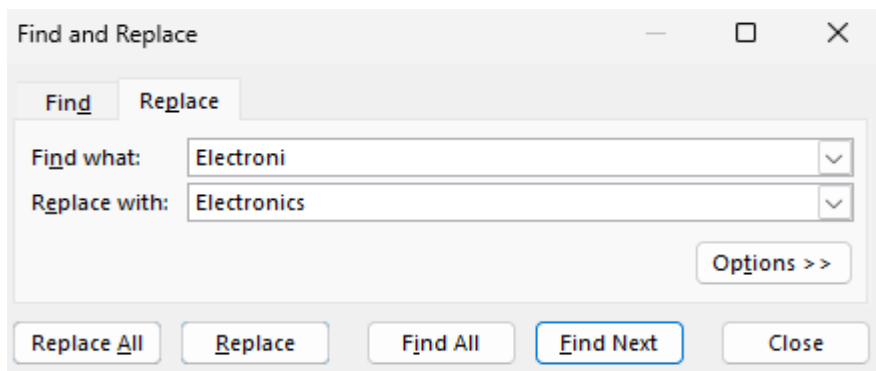
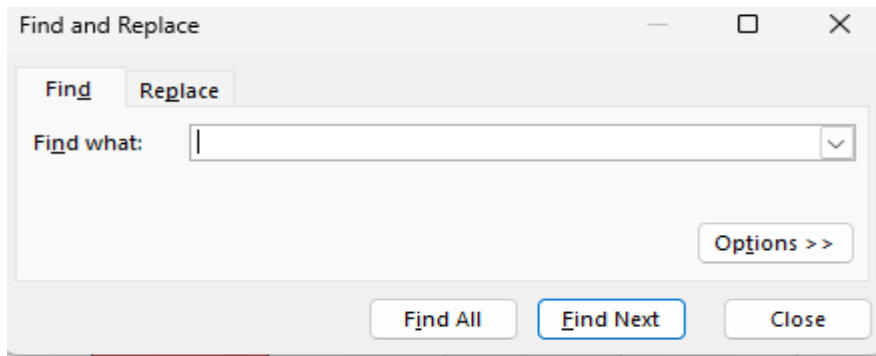
- ☒ (Select All)
- ☒ Accessories
- ☒ Electroni
- ☒ Electronics
- ☒ Fashion
- ☒ Kitchen
- ☒ Outdoor
- ☒ (Blanks)

OK Cancel

Electroni
Electroni
Electroni
Electroni
Electroni

Task Comments	Output Values
Column and fix any typos or misspellings in the "Category" column.	Refer Bold Highlighted cells in Column L

- Sort the column F
- Find the misspelled category "Electroni"
- 3. Select Ctrl + F, Select Replace Tab
- 4. Type "Electroni" in Find What option and "Electronics" in Replace with option
- 5. Select Replace All
- 6. 5 values of cells contains word "Electroni" is replaced with word "Electronics"



3.4 Splitting and Merging Data:

- Split **Product ID** into: → **Manufacturing Date** (Column M)
- **Country Code** (Column N) using **Text to Columns**.
- Merged **Brand Name** and **Product Name** into a new column **Product Brand** (Column O).

Task Comments	Output Values
Split the "Product ID" column into two separate columns for "Manufacturing Date"	Refer to Column M

Formula Used
=LEFT(A2,2)

Task Comments	Output Values
Split the "Product ID" column into two separate columns for "Country Code"	Refer to Column N

Formula Used
=RIGHT(A2,2)

Task Comments	Output Values
Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".	Refer to Column O

Formula Used
= (C2) & " " & (B2)

3.5 Number Formatting:

- Formatted **Price** column to **Currency** using CTRL + SHIFT + 4.
- Formatted **Manufacturing Date** column to **DD-MM-YYYY**.

Task Comments	Output Values	Formula Used
Format the data type of the "Price" column to a currency format.	Refer Column J	CTRL+SHIFT+4

Task Comments	Output Values
Format the "Manufacturing Date" column to display dates	Refer Column P

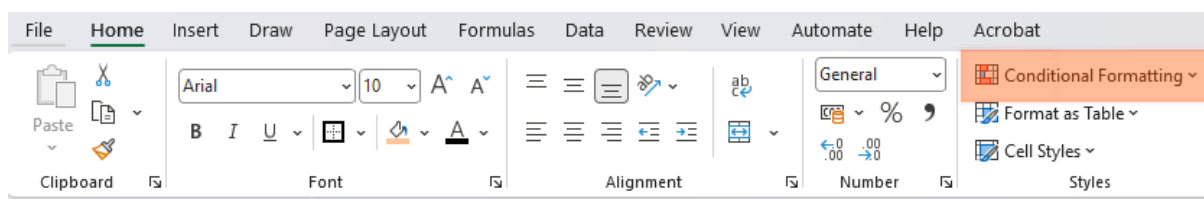
Formula Used
=TEXT(DATE(2025, MONTH(MID(A2,4,3) & " 1"), LEFT(A2,2)), "MM-DD-YYYY")

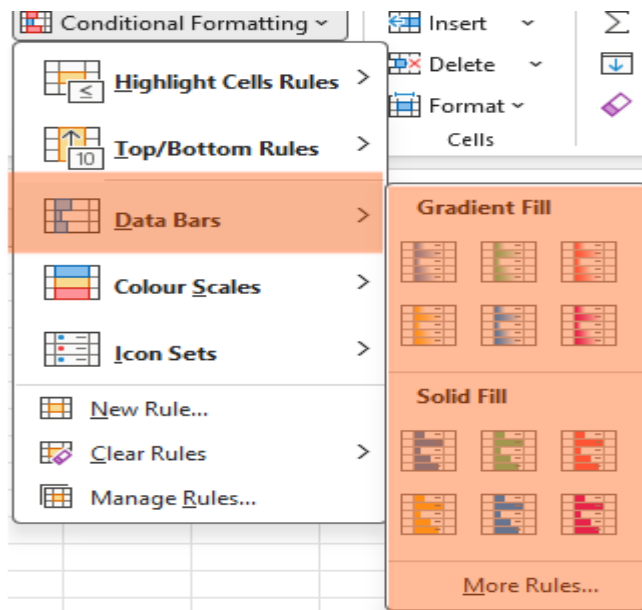
3.6 Conditional Formatting:

- Applied **Data Bars (Gradient Fill)** to the **Price** column.
- Highlighted all cells containing **"Electronics"** in the **Category** column using **Text That Contains** rule.

Task Comments	Output Values
Apply data bar or colour scales conditional formatting in the "Price" column.	Refer Column J

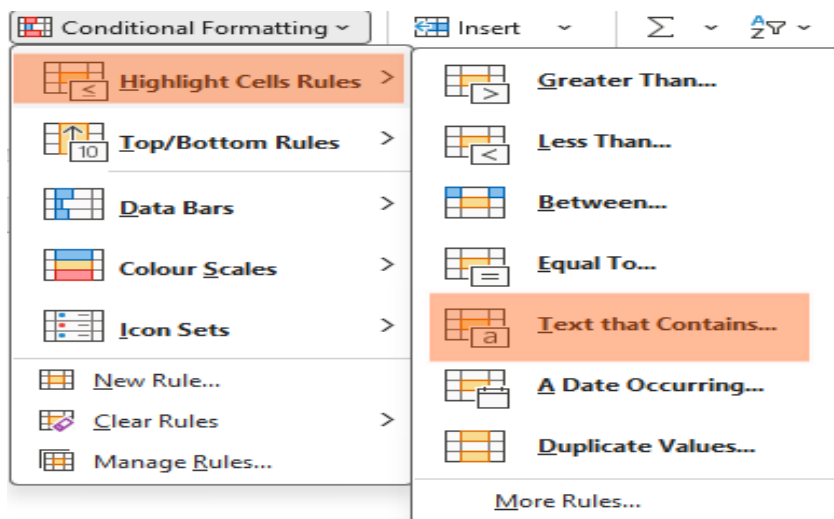
1. Select the column D
2. Select the option Conditional Formatting in the Column Styles
3. Select the Drop down --> Data Bars --> Gradient fills





Task Comments	Output Values
Conditional formatting in the "Category" column to highlight cells where the category is Electronics	Refer Column L

- Select the column F
- Select the option Conditional Formatting in Cell Styles
- 3. Select the Drop down --> Highlight Cells rules --> Text That contains --> Type "Electronics" --> Select Okay



Text That Contains
?
X

Format cells that contain the text:

Electronics

↑

with

Green Fill with Dark Green Text

▼

OK

Cancel

RAW DATA						OUTPUT VALUES									
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Manufacturing Date	Country Code	Product Brand	Manufacturing Date
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	09-JUL-FR	Watch	Casio	₹ 100.00	20	Accessories	9	FR	Casio Watch	07-09-2025
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	21-AUG-CA	Laptop Bag	Samsonite	₹ 50.00	35	Accessories	21	CA	Samsonite Laptop Bag	08-21-2025
09-JUL-FR	Watch	Casio	100	20	Accessories	07-JUN-UK	Headphones	Sony	₹ 297.86	45	Electronics	7	UK	Sony Headphones	06-07-2025
07-JUN-UK	Headphones	Sony		45	Electronics	16-APR-ES	Headphones	Bose	₹ 250.00	20	Electronics	16	ES	Bose Headphones	04-16-2025
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	18-FEB-CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	18	CA	Fitbit Smartwatch	02-18-2025
16-APR-ES	headphones	Bose	250	20	Electronics	20-AUG-CN	Smartwatch	Huawei	₹ 160.00	15	Electronics	20	CN	Huawei Smartwatch	08-20-2025
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	05-SEP-UK	Tablet	Apple	₹ 500.00	50	Electronics	5	UK	Apple Tablet	09-05-2025
16-APR-ES	headphones	Bose	250	20	Electronics	11-APR-US	Smartphone	Samsung	₹ 900.00	25	Electronics	11	US	Samsung Smartphone	04-11-2025
28-JAN-US	laptop	Dell	1000	30	Electronics	14-MAY-RU	Camera	Nikon	₹ 700.00	50	Electronics	14	RU	Nikon Camera	05-14-2025
11-APR-US	smartphone	Samsung	900	25	Electronics	17-JUN-IN	Laptop	Hp	₹ 950.00	25	Electronics	17	IN	Hp Laptop	06-17-2025
05-SEP-UK	Tablet	Apple	500	50	Electronics	24-MAY-CA	Fitness Tracker	Garmin	₹ 130.00	5	Electronics	24	CA	Garmin Fitness Tracker	05-24-2025
17-JUN-IN	Laptop	HP	950	25	Electronics	27-JAN-IT	Laptop	Asus	₹ 980.00	10	Electronics	27	IT	Asus Laptop	01-27-2025
27-JAN-IT	laptop	Asus	980	10	Electronics	28-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	28	US	Dell Laptop	01-28-2025
14-MAY-RU	Camera	Nikon	700	50	Electronics	29-SEP-CA	Smartphone	Google	₹ 800.00	45	Electronics	29	CA	Google Smartphone	09-29-2025
29-SEP-CA	Smartphone	Google	800	45	Electronics	01-MAR-UK	Sunglasses	Oakley	₹ 150.00	15	Fashion	1	UK	Oakley Sunglasses	03-01-2025
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	02-DEC-CA	Jeans	Levi'S	₹ 50.00	50	Fashion	2	CA	Levi'S Jeans	12-02-2025
17-JUN-IN	Laptop	HP	950	25	Electronics	03-JUN-CA	Sunglasses	Ray-Ban	₹ 297.86	25	Fashion	3	CA	Ray-Ban Sunglasses	06-03-2025
15-FEB-US	Sneakers	Nike	80	15	Fashion	07-MAR-CA	Dress	Zara	₹ 60.00	30	Fashion	7	CA	Zara Dress	03-07-2025
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	15-FEB-US	Sneakers	Nike	₹ 80.00	15	Fashion	15	US	Nike Sneakers	02-15-2025
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	19-JUL-UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	19	UK	Adidas T-Shirt	07-19-2025
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	03-MAR-US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	3	US	Keurig Coffee Maker	03-03-2025
07-MAR-CA	Dress	Zara	60	30	Fashion	09-JAN-CA	Microwave	Panasonic	₹ 80.00	20	Kitchen	9	CA	Panasonic Microwave	01-09-2025
02-DEC-CA	Jeans	Levi'S	50	50	Fashion	11-JUL-CA	Blender	Vitamix	₹ 400.00	40	Kitchen	11	CA	Vitamix Blender	07-11-2025
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	13-APR-CA	Toaster	Hamilton	₹ 40.00	10	Kitchen	13	CA	Hamilton Toaster	04-13-2025
23-AUG-UK	Blender	Ninja	90	35	Kitchen	23-AUG-UK	Blender	Ninja	₹ 90.00	35	Kitchen	23	UK	Ninja Blender	08-23-2025
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	14-AUG-US	Camping Tent	Coleman	₹ 297.86	10	Outdoor	14	US	Coleman Camping Tent	08-14-2025
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	14-OCT-UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	14	UK	Timberland Hiking Boots	10-14-2025
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	08-DEC-DE	Coffee Maker	Nespresso	₹ 120.00	35	Unknown	8	DE	Nespresso Coffee Maker	12-08-2025

Conclusion:

- This project successfully demonstrates the essential steps of data cleaning and preparation using Excel. By handling missing values, correcting inconsistencies, removing duplicates, restructuring data, and applying formatting rules, the dataset becomes accurate, consistent, and ready for analysis